

Lightweight Group Pre-Handover Authentication Scheme for Aviation 5G Air-To-Ground Networks

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Abstract—With the emergence of fifth-generation (5G) technology, the air-to-ground (ATG) communication system based on 5G public mobile communication technology makes it possible to provide data services for in-flight communication. Customer premises equipment (CPE) is connected to ATG base stations to provide network access for user devices in-cabin. However, the handover authentication performed by CPEs during a handover between ATG base stations poses security risks and incurs a lot of overhead. Furthermore, frequent handovers between base stations and the limited computational resources of the CPE make the protection of security properties even more challenging. In this paper, we propose a lightweight group pre-handover authentication scheme for aviation 5G ATG networks. By leveraging the predictability of flight paths, all CPEs on the same aircraft can complete handover authentication before arriving at the next base station, and handover authentication delay can be ignored, providing seamless communication services for user devices in the cabin. The scheme utilizes the extended Chebyshev chaotic map and symmetric encryption technology to reduce computational overhead and adapt to the limited computing resources of CPE. Security and performance analysis show that our scheme achieves lightweight and efficiency while ensuring security performance and outperforms other relevant schemes.

Index Terms—civil aviation, 5G Air-To-Ground, handover authentication, extended Chebyshev chaotic map

I. INTRODUCTION

As a typical high-speed airborne scenario, data transmission on commercial plane suffers from severe path loss, Doppler shift and other issues, leading to poor quality when passengers attempt to communicate with the ground and inability to access network services during the flight, but also interfere with normal plane navigation. Information isolation still exists in the air, despite the growing prevalence of mobile broadband services on land.

With the emergence of the fifth-generation (5G) technology, air-to-ground (ATG) communication system based on 5G wireless broadband technologies makes it possible to provide robust data services for in-cabin communications. The 5G ATG system deploys ground ultra-high-power ATG base stations within the airline to provide wireless signal coverage for planes. Onboard passenger wireless terminals are connected to the in-flight customer premises equipment (CPE), which then communicates with the ATG base station through the onboard CPE to connect the plane to the ground network.

While 5G ATG makes in-flight internet connectivity possible, the insecure access of CPE poses a significant risk to passengers' data security and privacy. Additionally, 5G networks use smaller cells than Long-Term Evolution (LTE), resulting in higher handover frequencies and greater authentication delays. Existing handover authentication schemes have limitations, with those that provide strong security guarantees often requiring significant computational costs or using complex cryptographic operations [1]–[4], which is unsuitable for the handover requirements in high-mobility scenarios. Apart from the longer handover time, these schemes are also incompatible with those deployed in existing communication networks, requiring additional expenses that limit their practicality [5].

In this paper, we propose a group pre-handover authentication scheme for aviation 5G ATG networks. All CPEs on the same plane form a CPE group. Software-defined networking generates a handover ticket (HT) for each CPE upon initial network access. During pre-handover authentication, CPEs use the HT to complete the authentication process based on aggregated message authentication codes and key negotiation based on extended Chebyshev chaotic maps.

Our proposed scheme aims to address the challenge of handover authentication in aviation 5G ATG networks. Specifically, it allows authentication and session key negotiation to be completed before handover, eliminating additional authentication delays. This is possible due to 5G ATG base stations being deployed along the flight route and commercial planes having a relatively fixed and predictable flight path. Furthermore, our scheme offers robust security features while minimizing overhead. To reduce computational overhead, we leverage extended Chebyshev chaotic maps instead of computationally expensive bilinear pairings or modulo exponentiation operations. Additionally, we aggregate the authentication and key agreement messages of CPEs to reduce signaling and communication overhead.

II. RELATED WORK

5G ATG networks are in the exploration phase but network security issues have rarely been studied. Significantly, some handover authentication schemes have been proposed for similar high mobility scenarios. The Third Generation Partnership Project (3GPP) has also developed standards for handover authentication [6], but they have many security vulnerabilities

and performance issues, such as multiple rounds of signaling message interaction leading to large handover delays, complex handover key management system, key forward secrecy failure, and lack of mutual authentication [7], [8]. Furthermore, the current 3GPP standard does not take group handover authentication into account.

For similar high-mobility scenarios, Kong et al. [9] proposed a secure authentication protocol for mobile relay group handover based on proxy re-encryption technique. The ciphertext can be decrypted by the target eNB without the involvement of the source MME. However, the scheme cannot provide perfect forward and backward security, and the exhaustive search consumes a significant amount of computational overhead. Haddad et al. [1] proposed a group handover authentication scheme in LTE-A networks for privacy-preserving MMEs. The user generates a signature with a one-time public-private key pair, and the MRN aggregates and forwards all signed messages to the eNB. Although it provides anonymous authentication and protects the user's privacy information, multiple bilinear mapping operations incur significant communication and computational overhead. Cao et al. [10] proposed a group pair routing handover authentication scheme based on trajectory prediction. MRNs and base stations on the running trajectory route are aggregated into groups separately and share the same group key, which improves the handover efficiency. However, it cannot achieve perfect forward security, backward security and unlinkability, etc.

Ma et al. [11] proposed two fixed trajectory group pre-handover authentication schemes in 5G high-speed railway networks. The first scheme has less overhead for handover, but cannot achieve anonymity and full forward secrecy, etc. The second scheme achieved higher security but has higher overhead due to cryptographic operations on elliptic curves. Singh et al. [2] proposed a handover protocol for group MRNs in LTE-A networks. The proposed method does not require the MME to be involved in the key agreement, which reduces the computational overhead and storage overhead of the MME. However, the authentication process requires modulo-exponential operations, which brings a large bandwidth consumption and computational overhead. Li et al. [12] proposed a group pre-handover authentication scheme based on a trusted MRN. Authentication is performed through a trusted execution environment deployed by the MRN, which improves the handover efficiency. However, this scheme requires MRNs to perform path prediction and deploy trusted execution rings in MRNs. In addition, they lack formal security analysis and overhead evaluation.

In addition, there are many handover authentication schemes for other network scenarios, such as handover authentication schemes for a group of UEs or machine type communication devices (MTCDs). Some of the proposed schemes are based on Security Context Transmission (SCT) [13]–[17]. Although they simplify the authentication process, they still require sending a large number of access request messages for multiple devices, incurring significant signaling overhead, and may lead to key leakage problems [1], [3], [4], [10], [11], [17]–[20]. However,

[4] uses multiple signatures and modulo power operations with high computational cost, and similarly [1] also uses modulo power operations. Yan et al. [3] proposed an efficient authentication protocol for all handover scenarios of vehicle queues. The identities of the base station and vehicle are mutually authenticated by certificate-free aggregated signature, which also reduces signaling overhead and does not have a key escrow problem, but the computation of bilinear maps is required for the first vehicle in the vehicle queue, which has a large computational overhead. Lai et al. [21] proposed an anonymous handover authentication scheme to support multi-user access, which provides lightweight user grouping and internal group authentication. Moreover, the use of AMAC with detection (AMAD) technique to authenticate multiple entities simultaneously is able to accurately identify and output a list of identities corresponding to malicious MACs.

III. PRELIMINARIES

In this section, we introduce the definitions of the Chebyshev chaotic map and extended Chebyshev polynomials, and the properties of them [22], [23].

Define 1 Chebyshev Chaotic Map

For $n \in \mathbb{N}$, $x \in [-1, 1]$, the Chebyshev polynomial is denoted by

$$T_n(x) = \cos(n \cdot \arccos(x)). \quad (1)$$

The recurrence of $T_n(x)$ is presented as follows.

$$T_n(x) = \begin{cases} 2xT_{n-1}(x) - T_{n-2}(x) & , n \geq 2 \\ x & , n = 1 \\ 1 & , n = 0 \end{cases} \quad (2)$$

Property 1 Semigroups Property

The semigroup property of the Chebyshev polynomials is defined as follows :

$$\begin{aligned} T_r(T_s(x)) &= \cos(r \cdot \cos^{-1}(\cos(s \cdot \cos^{-1}(x)))) \\ &= \cos(rs \cdot \cos^{-1}(x)) \\ &= T_{rs}(x) \end{aligned} \quad (3)$$

where $x \in [-1, 1]$, r and s are positive integers.

Property 2 Commutative Property

The commutative property of the Chebyshev polynomials under composition is presented: $T_r(T_s(x)) = T_s T_r(x)$

Define 2 Extended Chebyshev Polynomials

For the extended Chebyshev polynomials defined on $(-\infty, +\infty)$, the stability is improved while the semigroup properties are retained. The definition is as follows.

$$T_r(T_s(x)) \bmod p = T_{rs}(x) \bmod p = T_s(T_r(x)) \bmod p \quad (4)$$

where p is a large prime and $n \geq 2$.

Property 3 Chaotic Maps-Based Discrete Logarithm (CMDL) Problem

For any given x , $T_s(x) \bmod p$, cannot be computed in any polynomial time.

Property 4 Chaotic Maps-Based Diffie-Hellman(CMDH) problem

For any given $x, T_r(x) \bmod p, T_s(x) \bmod p, T_{rs}(x) \bmod p$ cannot be computed in any polynomial time.

IV. SYSTEM MODEL

The network architecture of the 5G ATG communication system consists of four main components, namely SDN controller [24], ATG base station, CPE, and UE. SDN controller is simply a program running on a server that can be placed anywhere in the 5G core network [11], [25]. We use SDN as a platform for user authentication, authorization, and mobility management [24]. SDN and the ATG are connected via wired links, while the ATG base station and the CPEs are connected by wireless links. The on-board CPE performs initial authentication and handover procedures and then acts as a base station to provide uninterrupted connectivity to the UE. ATG is a ground-based specialized base station deployed along air routes.

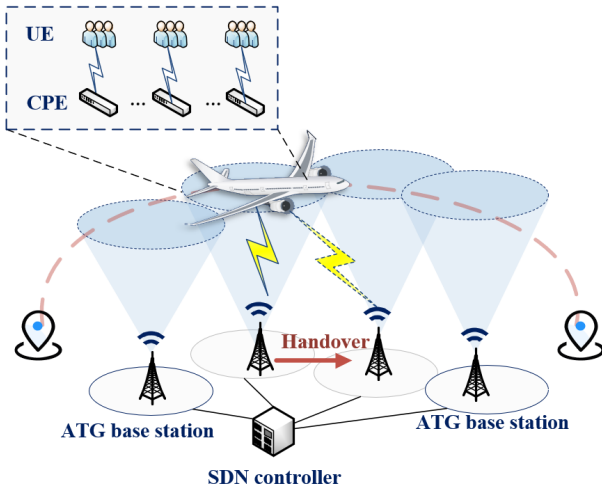


Fig. 1. The Network Architecture of the 5G ATG

V. PROPOSED SCHEME

Our proposed scheme consists of two phases: the initial authentication phase and the pre-handover authentication phase. At the initial authentication phase, CPEs in the same cabin form a CPE group. SDN creates an HT for a CPE when it connects for the first time to the network. With the help of SDN, the CPE and ATG base station accomplish mutual authentication and the session key agreement. When the CPE group is about to leave one base station and move to another, the CPE group performs the pre-handover authentication procedure. Each CPE encrypts its privacy information and generates a MAC for it. The MRN group leader aggregates all the information and sends it to SDN for verification. If the verification is successful, SDN chooses the suitable next ATG station and assists it to negotiate the session key with the CPE group members. In this way, the handover authentication delay of the CPE is reduced, ensuring seamless passenger communication services.

A. System Setup

The SDN controller selects a large prime number q , and $\theta \in_R (-\infty, +\infty)$. Besides, the SDN controller picks $sk_{SDN} \in_R Z_q^*$ as the master secret key, pk_{SDN} as the raw public key, where $pk_{SDN} = T_{sk_{SDN}}(\theta)$. Subsequently, it Choose one-way secure hash functions H . Finally, public $\{q, H, pk_{SDN}\}$.

B. Initial Authentication Phase

When each CPE_i connects to the 5G ATG network firstly, it should perform the initial authentication with SDN controller. We have adopted the short signature [26] in the phase.

- 1) $CPE_i \rightarrow ATG_1 : (C_i, Z_i)$
 - a) Choose $z_i \in_R Z_q^*$, compute $Z_i = T_{z_i}(\theta), U_i = T_{z_i}(pk_{SDN}) \bmod q$
 - b) Compute $C_i = H(U_i) \oplus ID_{CPE_i}$, where ID_{CPE_i} is the identity associated with CPE_i .
- 2) $ATG_1 \rightarrow SDN : (C_i, Z_i)$
- 3) $SDN \rightarrow ATG_1 : (A_i, D_i, s_i, ID'_{CPE_i}, K_{1i}, Y_i)$
 - a) Compute $U'_i = T_{sk_{SDN}}(Z_i)$, and decrypt $ID'_{CPE_i} = H_1(U'_i) \oplus C_i$
 - b) Choose $y_i \in_R Z_q^*$, compute $Y_i = T_{y_i}(\theta)$, $K_{1i} = T_{y_i}(Z_i)$
 - c) Compute $P_i = T_{sk_{SDN}}(Z_i \parallel ID'_{CPE_i})$
 - d) Choose $r_i \in_R Z_q^*$, compute $B_i = H(P_i)$, $k = B * T_{r_i}(\theta)$, $D_i = H(P_i, k)$, $s_i = r_i - D_i * sk_{SDN}$
 - e) Compute $A_i = H(U'_i) \oplus P_i$, and store handover ticket(ID', P_i)
- 4) $ATG_1 \rightarrow CPE_i : (A_i, D_i, s_i, Y_i)$
 ATG_1 stores K_{1i} as the session key, takes ID'_{CPE_i} as the identity of D'_i , and forwards (A_i, D_i, s_i, Y_i) to CPE_i .
- 5) $CPE_i \rightarrow ATG_1 : \text{Authentication Confirm Message}$
 - a) CPE_i decrypt $P'_i = H(K_{1i}) \oplus A_i$
 - b) Compute $B_i = H(P'_i)$, $k' = B'_i * T_D(pk_{SDN}) * T_{s_i}(\theta)$ and $D'_i = H(P_i, k')$ and verify if $D = D'$.
 - c) If it is, CPE_i compute the session key $K_{1i} = T_{z_i}(Y_i)$, stores the handover ticket P_i , and sends a Authentication Confirm Message to ATG_1 .

After the initial authentication phase, the session key K_{1i} can be employed to ensure the security of the communication channel between the ATG_1 and the CPE_i and the handover ticket P_i can be used to accomplish the pre-handover authentication phase.

C. Pre-Handover Authentication Phase

When the CPE detects that a handover is going to occur, the following procedure is performed.

- 1) $CPE_1 \rightarrow CPE_i$: Broadcast a pre-handover request message to inform all group members to start the pre-handover authentication process.
- 2) $CPE_i \rightarrow CPE_1 : (ID_{ATG_1}, X_i, E_{1i}, MAC_{1i})$
 - a) Choose $x_i \in_R Z_q^*$, compute $X_i = T_{x_i}(\theta), K_i = T_{x_i}(pk_{SDN})$

- b) Compute $MAC_{1i} = H(P_i, ID_{CPE_i}, ID_{ATG_1}, M_i, X_i)$, where M_i is necessary information for certain handover, such as navigation plans. Encrypt and obtain $E_{1i} = Enc_{K_i} \{ID_{CPE_i} \| M_i\}$
- c) Send $(ID_{ATG_1}, X_i, E_{1i}, MAC_{1i})$ to CPE_1
- 3) $CPE_1 \rightarrow ATG_1: ((X_i, E_{1i})_{i=1,\dots,n}, MAC_1)$
 - a) After receiving messages from other members of the group, the same operation as CPE_i is performed, but when computing E_{11} , $E_{11} = Enc_{K_1} \{M' \| ID_{CPE_i} \| M_i\}$, where M' is the necessary information for handover, such as description of the member, current location information.
 - b) In addition, CPE_1 has to compute $MAC_1 = MAC_{1i} \oplus \dots \oplus MAC_{in}$
- 4) $ATG_1 \rightarrow SDN: (ID_{ATG_1}, ((X_i, E_{1i})_{i=1,\dots,n}, MAC_1))$
After receiving the pre-handover request from the user, ATG_1 forwards the message to the SDN over the established secure channel.
- 5) $SDN \rightarrow ATG_2: ((X_i, M'_i)_{i=1,\dots,n})$
 - a) Compute $K_i = T_{sk_{SDN}}(X_i)$, and decrypt E_{1i} to obtain ID_{CPE_i}, M_i , and search the handover ticket P_i from the database by using ID_{CPE_i}
 - b) Verify MAC_1 by using P_i . If the authentication is successful, SDN selects the appropriate next base station ATG_2 for the CPE group based on the existing path information and navigation information, and sends a handover notification through the secure channel that has been established in advance.
- 6) $ATG_2 \rightarrow SDN: (R, MAC_2, M_{ATG_2})$
 - a) Choose $r \in_R Z_q^*$, $R = T_r(\theta)$, and compute the session key $K_{2i} = T_r(X_i)$ for each CPE
 - b) Compute $H(K_{2i}, M'_i)$ and aggregate $MAC_2 = H(K_{21}, M'_1) \oplus \dots \oplus H(K_{2n}, M'_n)$
 - c) M_{ATG_2} is some information about handover.
- 7) $SDN \rightarrow ATG_1: (R, M_{ATG_2}, HV)$
Compute $HV = H(R, M_{ATG_2}, MAC_2)$ by using MAC_2 .
- 8) $ATG_1 \rightarrow CPE_1: (R, M_{ATG_2}, HV)$
- 9) $CPE_1 \rightarrow CPE_i: \text{Broadcast } R$
- 10) $CPE_i \rightarrow CPE_1: ACK_i$
Compute session key $K_{2i} = T_{x_i}(R)$ and $ACK_i = H(K_{2i}, M_i)$
- 11) After receiving all messages, CPE_1 computes $ACK = ACK_1 \oplus \dots \oplus ACK_n$, $HV' = H(R, M_{ATG_2}, ACK)$. If the verification is successful, CPE_1 sends a pre-handover completion message to all group members and SDN .

When the plane enters the range of the target station ATG_2 , CPE_i can establish communication immediately with the new session key K_{2i} .

VI. SECURITY ANALYSIS

We mainly analyse the security of pre-handover authentication phase. Our proposed scheme consists of two phases, and security properties of the initial authentication phase can be enhanced in some physical ways. Therefore, the main factor affecting the user communication security is the pre-handover authentication phase as it occurs during the cruising phase.

Mutual Authentication: Each CPE_i uses the handover ticket to generate MAC_{1i} , and the group leader aggregates all MACs. Only the legitimate CPE group can generate a valid MAC , if there is an invalid MAC , the authentication will fail; Since each CPE_i encrypts its privacy information M_i using the K_i , only the legitimate SDN can obtain M_i and generate a valid hash HV . Therefore the CPE group completes the verification of SDN by verifying the hash.

Key Agreement: The session key is computed by the equation $K_{2i} = T_r(X_i) = T_{x_i}(R)$, where computing the random number x_i or r is equivalent to solving the Chaotic Map Discrete Logarithm Problem (CMDL) or the Chaotic Map Diffie-Hellman Problem (CMDH).

Anonymity and Unlinkability: The privacy message M_i of CPE_i is encrypted using K_i , so that the scheme can satisfy anonymity. In addition, using random numbers X_i in each message, it is impossible to determine whether two messages are computed by the same CPE group members. Therefore, this scheme can achieve anonymity and unlinkability.

Perfect Forward/Backward Secrecy (PFS/PBS): The session key K_{2i} is obtained by equating $T_{x_i}(R)$, even if the handover ticket of CPE_i is compromised, the adversary cannot obtain the random number x_i through X_i , because it is equivalent to solving the chaotic Map Based Discrete Logarithm problem (CMDL), so the adversary cannot compute the session key. Similarly, even if the private key of the SDN is compromised, the adversary cannot obtain the random number r and cannot further compute the session key.

Forward/Backward Key Separation (FKS/BKS): Since the random numbers x_i and r are used to compute the session key, each key is independent of the other. Even if the current session key is derived, no adversary can obtain the previous or next session key.

Resistance against Replay Attacks: Replay attacks can be resisted by using random numbers x_i and r .

TABLE I
SIGNALING OVERHEAD FOR n CPE

Scheme	Signaling Overhead
Our scheme	6
FTGPHA2 [11]	9
5G standards [6]	$8n$
Haddad [1]	9
EGHA [3]	$2n + 5$

Resistance against Impersonation Attacks: On the one hand, only the legitimate CPE_i holds the handover ticket P_i and it is not directly transmitted during the handover

TABLE II
COMPUTATIONAL OVERHEAD FOR n CPE

Scheme	CPE group	ATG and SDN	Total
Our scheme	$3nT_c + (2n+1)T_H + nT_E = 5.48n + 1.21$	$(2n+1)T_c + (2n+1)T_H + nT_E = 4.81n + 1.88$	$10.29n + 3.02$
FTGPHA2 [11]	$3nT_M + (3n+1)T_H + T_E$ $= 1.5 \times 10^3 n + 2.26$	$(2n+4)T_M + (4n+1)T_H + (2n+1)T_A + T_E$ $= (n+2) \times 10^3$ $4nT_H = 4.84n$	$(2.5n+2) \times 10^3$
5G standards [6]	$4nT_H = 4.84n$	$4nT_H = 4.84n$	$9.68n$
Haddad [1]	$(3n+1)T_P + 2nT_S + 4nT_H + nT_M$ $= (2.76n + 0.84) \times 10^4$	$(n+1)T_P + nT_S + nT_H$ $= (9.36n + 8.36) \times 10^3$	$(3.7n + 1.68) \times 10^4$
EGHA [3]	$(9n+7)T_H + (2n+1)T_A + 3nT_M$ $+5T_P + T_{E/D} = 1.5 \times 10^3 n + 4.18 \times 10^4$	$(2n+11)T_H + (3n+1)T_A + (n+1)T_M$ $+5T_P + T_{E/D} = 0.51 \times 10^3 n + 4.18 \times 10^4$	$2.01 \times 10^3 n + 8.36 \times 10^4$

process, and any adversary without a valid handover ticket cannot forge it; the private information M_i of CPE_i is encrypted using K_1 , and any adversary without a valid sk_{SDN} can't further generate the correct hash HV ; As mentioned above, no adversary can compute the session key and thus is able to resist the counterfeit attack.

VII. PERFORMANCE ANALYSIS

In this section, we mainly compare our proposed scheme's performance with 5G standards [6] and these closely related schemes in [1], [11] and [3]. Our proposed scheme consists of an initial authentication phase and a pre-handover authentication phase. The initial authentication phase occurs before the takeoff of the plane, while the pre-handover authentication phase occurs during the cruising phase which typically consumes the majority of a flight. As the pre-handover authentication phase is the primary determinant of user communication quality in this scenario, we have focused our analysis associated with it.

A. Signaling Overhead

We analyse the potential for network congestion by analyzing signaling costs, which we quantified in terms of the number of signaling messages exchanged between n CPEs and the 5G ATG network, excluding messages within the CPE group.

From Table I, it can be seen that our scheme has a constant signaling overhead, and significantly lower signaling overhead than the 5G standards [6] and EGHA [3]. Our scheme has a similar performance with the FTGPHA [11] and Haddad [1] because group members are not required to exchange information with the base stations directly.

TABLE III
EXECUTION TIME OF CRYPTOGRAPHIC ELEMENTS

T_C	T_M	T_A	T_H	$T_{E/D}$	T_P	T_S
0.67	0.50×10^3	1.39	1.21	1.05	8.36×10^3	1.00×10^3

B. Computational Overhead

Only a Chebyshev polynomial operation T_C , an elliptic curve point addition operation T_A , an elliptic curve scalar

multiplication operation T_M , a modular exponentiation operation T_S , a bilinear pairing operation T_P , a symmetric encryption/decryption operation $T_{E/D}$, and a one-way hash operation T_H are considered in the computation overhead analysis, while other operations with lower time complexity are neglected. We compare the computational costs of cryptographic operations by referring to the simulation time of previous studies [11], [27] using the OpenSSL library on an Intel(R) Core(TM) i7-7500U CPU with a clock speed of 2.70 GHz in Table III. We do not distinguish the computing power between CPE, ATG base station, and SDN.

Table II shows the computational overhead for n CPE. Our proposed scheme has demonstrated comparable computational overheads to the 5G standards [6], and significantly lower overheads than other schemes [1], [3], [11]. Our proposed scheme mainly involves Chebyshev polynomial operations, symmetric encryption/decryption operations, and one-way hash operations, resulting in relatively low computational overheads. In contrast, Scheme [11] incurs a larger computational overhead due to the use of elliptic curve cryptography operations, while Schemes [1], [3] use bilinear pairing operations resulting in significant overheads. Even though the 5G standards have the lowest computational overhead, they are vulnerable to attacks due to security vulnerabilities.

TABLE IV
COMMUNICATION OVERHEAD FOR n CPE

Scheme	Communication Overhead (bytes)
Our scheme	$272n + 164$
FTGPHA2 [11]	$368n + 560$
5G standards [6]	$80n$
Haddad [1]	$1572n + 440$
EGHA [3]	$643n + 4384$

C. Communication Overhead

According to the NIST standard, the security levels in the compared schemes are all equivalent to AES128 bits, the block length and key size of symmetric encryption and decryption algorithms are 128 bits, the key length of elliptic curve based cryptography algorithm is 256 bits, and the public key length of finite field based cryptography algorithm is 3072 bits and the private key is 256 bits. In addition, the output length of hash, random number, Chebyshev chaotic map length,

and information related to transmission are 128 bits, and the timestamp is 32 bits.

In terms of communication overhead, we mainly analyze the total size of the messages sent by n CPEs. Table IV shows our scheme has a communication overhead similar to [11], smaller than that of schemes [1], [3], and larger than 5G standards [6]. Since 5G standards [6] transmitted messages mainly consist of random numbers and the output of hash operations, the overhead of communicating is smaller than our scheme, while schemes [1], [3] rely on elliptic curve cryptography or finite field cryptography, which leads to relatively higher communication overhead. Our proposed scheme achieves more security properties than [1], [6].

VIII. CONCLUSION

In this paper, we proposed a lightweight group pre-handover authentication scheme for the handover requirements of aviation 5G ATG scenarios. We aggregate authentication messages to achieve pre-handover authentication. Additionally, with the assistance of SDN, the CPE group and the target ATG base station can accomplish the handover authentication before the CPE arrives in advance, due to aviation planes having a relatively fixed and predictable flight path. Therefore, the handover delay could be ignored, greatly improving the handover efficiency and ensuring passengers' seamless communication experiences. The security and performance analysis demonstrates that our scheme is excellent in guaranteeing security properties and achieves lightweight and ideal efficiency simultaneously, which is very suitable for aviation 5G ATG networks.

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